

Features

- Supports 0-10V/PWM/Rx dimming
- THD <10%
- Output current adjustable via DIP switch
- Surge: L-N 2kV
- Flicker free
- 1% dimming depth
- IP20
- AUX: 12Vdc&200mA; ripple voltage<80mV
- Suitable for Class II light fixtures
- 5-year warranty (please refer to the warranty condition)



Applications

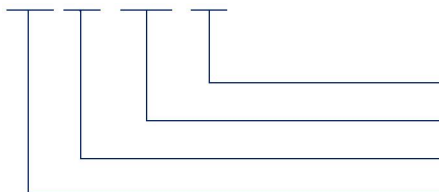
· Indoor office lighting · decorative lighting · commercial lighting · residential lighting

Descriptions

LF-ABA040A-1050-42 is a constant current LED driver with the maximum output power of 40W. Its rated input voltage ranges from 220 to 240Vac and its output current can be adjusted via DIP switch from 800mA to 1050mA with every 50mA as a step. It supports 12V AUX that can connect to external smart modules. Besides, it has all-round protection, including open circuit protection and short circuit protection. It is suitable for panel lights, downlights and so on.

Product Model

LF - ABA 040A - 1050 - 42



- 42: maximum output voltage: 42V
- 1050: maximum output current: 1050mA
- 040: rated power: 40W; A: 12V AUX
- ABA: indoor 3-in-1 dimming LED driver series

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■ Electrical Characteristics

Model		LF-ABA040A-1050-42					
Output	Output Voltage	25-42V				25-40V	25-38V
	Output Current	800mA	850mA	900mA	950mA	1000mA	1050mA
	Ripple current (≤100Hz)	± 3.3%					
	Flicker Index	IEC-Pst≤1, CIE SVM≤0.4 According to IEEE Std 1789-2015.					
	Current Tolerance	± 5%					
	Temperature Drift	± 10%					
	Startup Time	<1S					
	AUX	12± 1.5V&200mA					
	AUX Ripple Voltage	<80mV					
Input	Rated Input Voltage	220-240Vac					
	Input Voltage Range	198-264Vac					
	Input Frequency	50/60Hz					
	Input Current	0.3A max.					
	PF	≥0.97					
	THD	<10%					
	Efficiency	≥88%					
	Inrush Current	<40A&250uS					
	Loading Quantity on Circuit Breaker	Model	B10	C10	B16	C16	
		Quantity (pcs)	22	22	35	35	
	Leakage Current	<0.7mA					
	Standby Power Consumption	≤0.5W (dims to off)					
Protections	Open Circuit	<59V					
	Short Circuit	Hiccup mode (auto-recovery)					
Environment Descriptions	Operating Temperature	-30℃ - +45℃					
	Operating Humidity	20-90%RH (without condensation)					
	Storage Temperature/ Humidity	-40℃~+80℃ (6 months in Class I environment); 10-90%RH (without condensation)					
	Atmospheric Pressure	86-106kPa					
Surge	L-N	2kV					

■ Electrical Characteristics

Safety and EMC	Certifications	ENEC, CE, RCM, CCC, UKCA
	Withstanding Voltage	I/P-O/P: 3.75kV&5mA&60S; I/P-DIM O/P-DIM: 1.5kV&5mA&60S
	Insulation Resistance	I/P-O/P I/P-DIM O/P-DIM: >100MΩ@500VDC
	Safety Standards	ENEC: EN61347-1:2015, EN 61347-2-13:2014/A1:2017, EN 62384:2020 CE-LVD: EN 61347-2-13:2014/A1:2017, EN 61347-1:2015, EN 62493:2015 RCM: AS 61347.2-13:2018 SAA: AS 61347.1:2016+A1:2018 AS 61347.2.13:2018 CCC: GB19510.1-2009, GB19510.14-2009 UKCA: BS EN IEC 55015:2019+A11:2020, BS EN 61547:2009, BS EN IEC 61000-3-2:2019, BS EN 61000-3-3:2013/A2:2021
	EMI	CE-EMC/RCM: EN55015, EN61000-3-2, EN61000-3-3 CCC: GB/T17743, GB17625.1, GB17625.2
	EMS	CE-EMC/RCM: EN61000-4-2, 3, 4, 5, 6, 11 CCC: GB/T17626.2, 3, 4, 5, 6, 11
Other Parameters	IP Rating	IP20
	RoHS	RoHS 2.0 (EU) 2015/863
	Tc Max	90°C
	Warranty Condition	5 years (Tc ≤82°C)
	Noise Level	≤25dB (this data is measured in a soundproof room and the noise collector should be 10CM away from LED driver)
Testing Equipment	AC power source: CHROMA6530, digital power meter: CHROMA66202, oscilloscope: Tektronix DPO3014, DC electronic load: M9712B, LED board, constant temperature and humidity chamber, lightning surge generator: Everfine EMS61000-5B, rapid group pulse generator: Everfine EMS61000-4A, spectroanalyzer: KH3935, hi-pot tester: EEC SE7440, flicker tester (flicker-free coefficient test) Everfine LFA-3000, etc.	
Test Remark	If there are no special remarks, the above parameters are tested at the ambient temperature of 25°C, humidity of 50%, maximum output power and input voltage of 230Vac/50Hz.	
Additional Remarks	1. It is recommended that user install over voltage protection, under voltage protection and surge protection devices in the power supply circuits of light fixtures to ensure electricity safety. 2. The LED driver used in combination with the end device is one of the accessories of the whole light fixture, and the EMC of the whole light fixture is not only susceptible to the driver itself, but to the LED light fixture and the whole light fixture's wiring. Thus, the manufacturer of LED light fixture should re-confirm the EMC of the whole light fixture before the whole light fixture is finished. 3. The number of LED drivers that can be connected to a circuit breaker and the inrush current are tested under the same conditions. 4. Lifud reserves the right to interpret any of the above parameters.	

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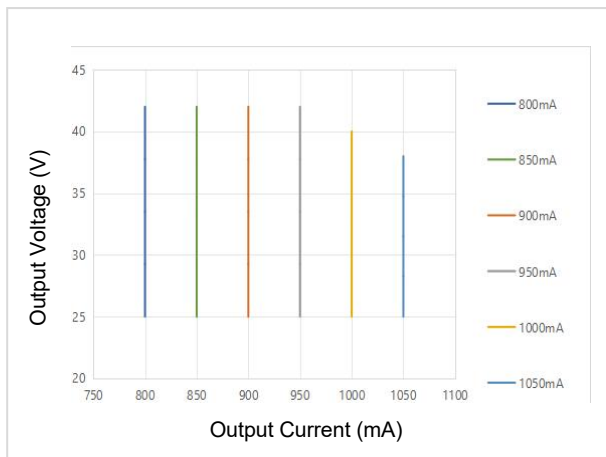
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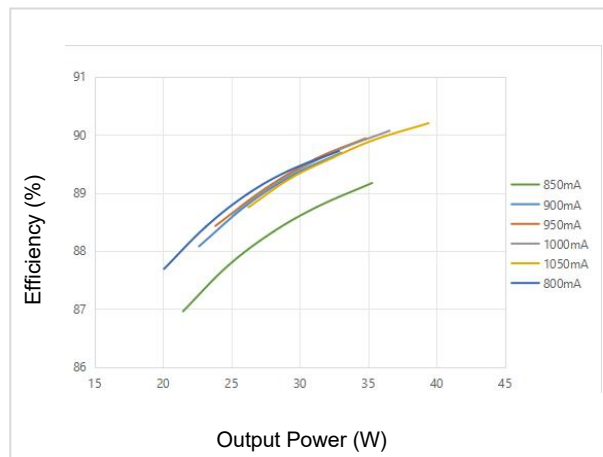
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■ Product Characteristic Curves

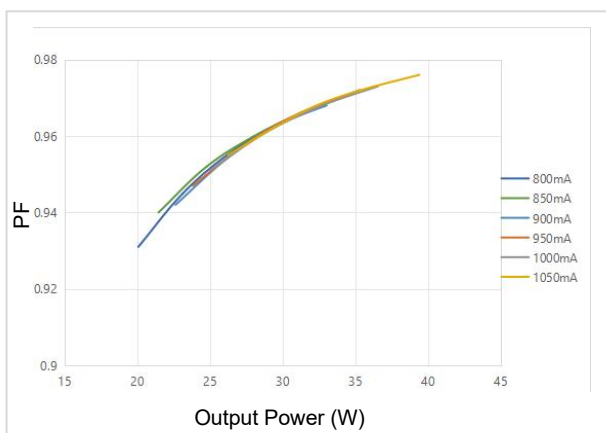
Working Window Curve



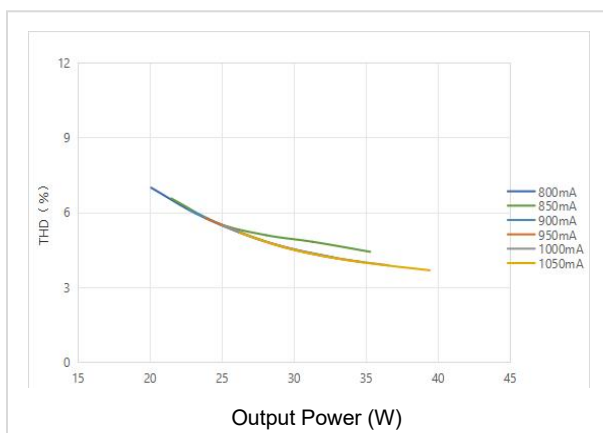
Efficiency Curve



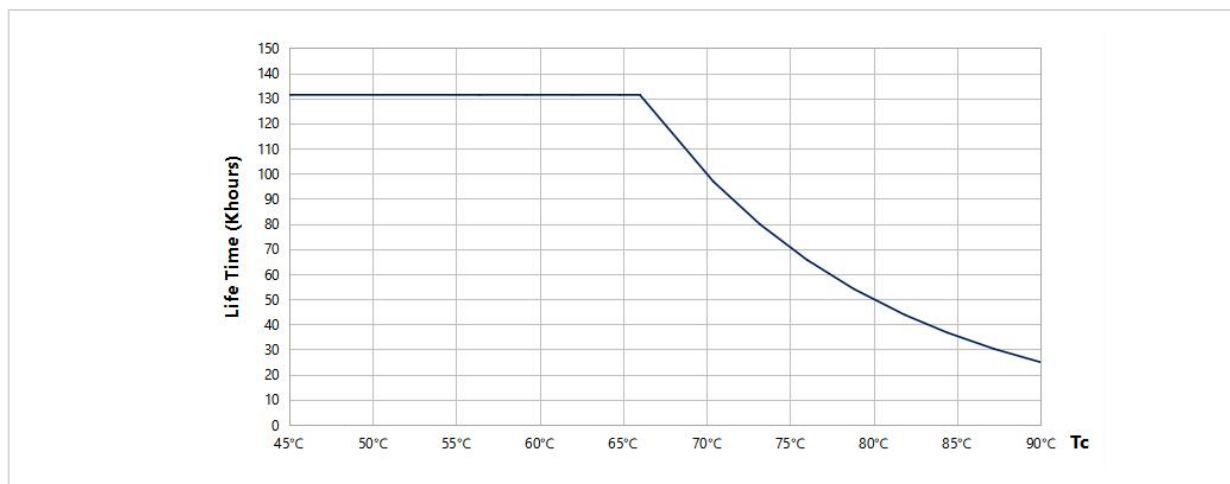
PF Curve



THD Curve



Lifetime Curve



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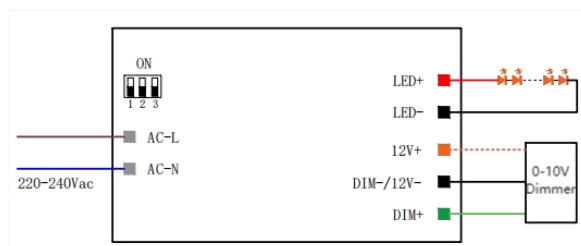
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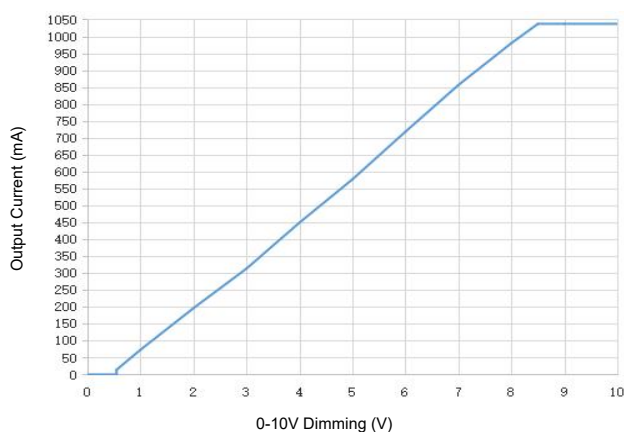
0-10V Dimming Operation

- Connect 0-10V signal to DIM terminal.
- In 0-10V dimming mode, when the input voltage is $0.55V \pm 0.05V$, the light turns on. When it's $0.35V \pm 0.1V$, the light turns off @ $U_o \text{ max} \& T_a = 25^\circ\text{C}$.
- Dimming depth: 1%
- DIM+/- (without signal connected): 100% rated current output

Wiring Diagram of 0-10V Dimming



Dimming Curve of Dim-to-Off Version



Input: 230Vac, output: 38Vdc/1050mA (The data is measured by Lifud 0-10V dimmer and the chart is for reference only)

PWM Dimming Operation

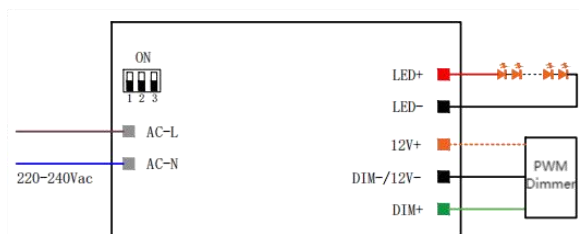
- Connect PWM signal to DIM terminal.
- Compatible signal range: 400-1000(Hz); amplitude: 9-10(V)

When it is $6\% \pm 1\%$, the light turns on;
when it is $3.5\% \pm 1\%$, the light turns off.

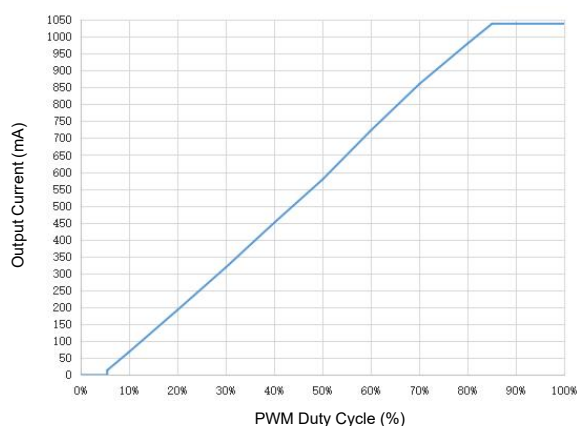
@ $U_o \text{ max} \& T_a = 25^\circ\text{C}$

- Dimming depth: 1%
- DIM+/- (without signal connected): 100% rated current output

Wiring Diagram of PWM Dimming



Dimming Curve of Dim-to-Off Version

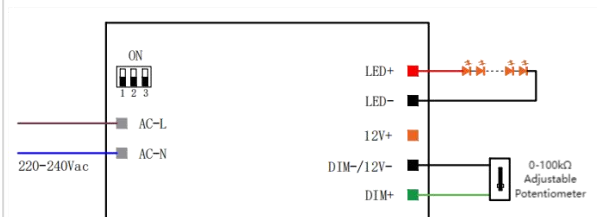


Input: 230Vac, output: 38Vdc/1050mA (The data is measured by PWM signal generator RIGOL and the chart is for reference only)

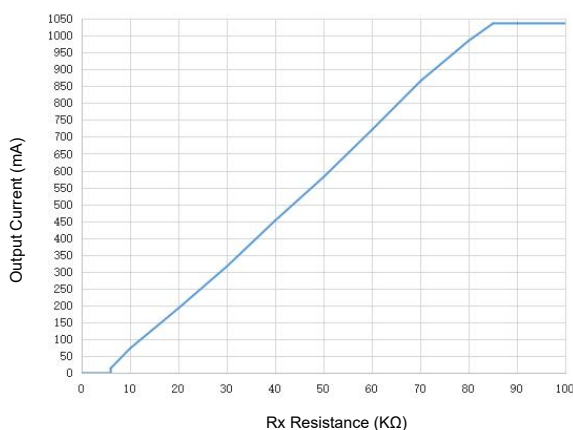
Rx Dimming Operation

- Connect Rx signal to DIM terminal.
- Range: 0-100K Ω @Uo max&Ta=25°C
When it is 5.5K $\Omega \pm 1$ K Ω , the light turns on;
when it is 3K $\Omega \pm 1$ K Ω , the light turns off.
- Dimming depth: 1%
- DIM+/- (without signal connected): 100% rated current output

Wiring Diagram of Rx Dimming



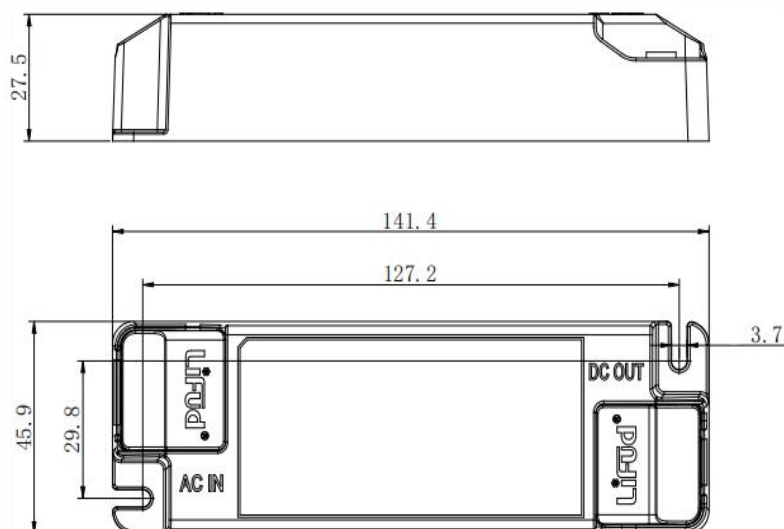
Dimming Curve of Dim-to-Off Version



Input: 230Vac, output: 38Vdc/1050mA (The data is measured by LEVITON dimmer and the chart is for reference only)

■ **Structure & Dimensions (unit: mm; tolerance: $\pm 0.5\text{mm}$)**

Model	Overall Appearance Dimension (L*W*H)	Distance Between 2 Positioning Holes (L*W)	Diameter of Positioning Hole (D)
LF-ABA040A-1050-42	141.4*45.9*27.5 mm ($\pm 0.5\text{mm}$)	127.2*29.8 mm ($\pm 0.5\text{mm}$)	3.7 mm



■ **Packaging Specifications**

Model	LF-ABA040A-1050-42
Carton Size	385*285*210 mm (L*W*H)
Quantity	10 pcs/layer; 7 layers/ctn; 70 pcs/ctn
Weight	0.12 kg $\pm 5\%$ /pc; 9.6 kg $\pm 5\%$ /ctn

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■ Transportation and Storage

1. Transportation

- Suitable transportation means: vehicles, boats and aeroplanes.
- In transit, it is necessary to prepare awnings for rain or sun protection. Moreover, please keep civilized loading and unloading to prevent the vibration or impact on LED driver as much as possible.

2. Storage

- The storage of LED driver shall conform to the standard of Class I environment.
- 1. When using the products which have been stored for more than 6 months, please re-test their electrical parameters firstly by powering on. Do not use them unless they are tested to be qualified.
- 2. When using the products which have been stored for 6 to 12 months, please run them on power for more than 2 hours firstly.
- 3. When using the products which have been stored for more than 12 months, please conduct an actual assessment of quality risks firstly. Direct use is not recommended.

Cautions

- Please use Lifud LED driver according to its parameters in the specification, otherwise the LED driver may malfunction.
- Using any incompatible light fixtures or those that have not been certified may cause fire, explosion or other risks.
- Man-made damage is beyond the scope of Lifud warranty service.

Remark: Lifud Tecnology Co., Ltd. reserves the right to interpret any content of this specification.